

Sensory Physiology

Human Physiology - Chapter 10

Biology Teaching Resources

December 24, 2025

Outline

10.1 Characteristics of Sensory Receptors

10.2 Cutaneous Sensations

10.3 Taste and Smell

10.4 Vestibular Apparatus and Equilibrium

10.5 The Ears and Hearing

10.6 The Eyes and Vision

10.7 Retina

10.8 Neural Processing of Visual Information

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Categories of Sensory Receptors

- **Sensory receptors** (感觉受体) are specialized structures that detect environmental stimuli
- Four main types based on stimulus:
 - **Mechanoreceptors** (机械感受器): Detect mechanical forces
 - **Chemoreceptors** (化学感受器): Detect chemical signals
 - **Photoreceptors** (光感受器): Detect light
 - **Nociceptors** (痛觉感受器): Detect tissue damage
- Receptors can be **exteroceptors** (外感受器) or **interoceptors** (内感受器)

Phasic and Tonic Receptors

- **Phasic receptors** (相位感受器) respond rapidly to stimulus onset but quickly adapt
- **Tonic receptors** (紧张性感受器) maintain steady firing as long as stimulus persists
- **Sensory adaptation** (感觉适应) allows us to ignore constant stimuli

Phasic and Tonic Receptors

- Comparison of receptor types:
- (a) **Phasic receptors**: Burst at stimulus onset, then rapid adaptation
- (b) **Tonic receptors**: Sustained firing during continuous stimulation
- Examples: Pacinian corpuscles are phasic; muscle spindles are tonic



Generator Potential

- **Generator potential** (生成电位) is a graded depolarization in sensory endings
- Similar to **EPSPs** (兴奋性突触后电位) but occurs in receptors
- When threshold is reached, action potentials are triggered

Generator Potential

- Sensory stimuli create local graded potentials
- Increasing stimulus strength produces larger generator potentials
- When generator potential reaches threshold (number 5), action potentials fire
- The magnitude encodes stimulus intensity



Generator Potential

- In **tonic receptors**, stimulus intensity is encoded by action potential frequency
- Stronger stimuli produce higher frequency firing
- This **frequency code** (频率编码) represents stimulus strength



Law of Specific Nerve Energies

- The **law of specific nerve energies** (特定神经能量定律) states each sensory nerve produces only one sensation
- **Adequate stimulus** (适当刺激): The stimulus requiring least energy to activate a receptor
- Example: Light is the adequate stimulus for photoreceptors
- **Paradoxical cold** (矛盾性冷觉): Hot stimuli can activate cold receptors through injury

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Receptive Fields and Sensory Acuity

- **Receptive field** (感受野): The area of skin that activates a sensory neuron
- **Sensory acuity** (感觉敏锐度) depends on receptive field size
- Smaller fields = greater acuity
- Fingertips have small fields (high acuity); back has large fields (low acuity)

Receptive Fields and Sensory Acuity

- **Cutaneous sensory receptors** (皮肤感觉受体) include various specialized structures
- **Free nerve endings** (游离神经末梢) detect pain, temperature, and light touch
- **Encapsulated receptors** (包裹性受体) include:
 - **Pacinian corpuscles** (环层小体): Deep pressure
 - **Meissner's corpuscles** (迈斯纳小体): Light touch and texture



Receptive Fields and Sensory Acuity

- **Two-point touch threshold test** (两点触觉阈值测试) measures acuity
- Two points stimulating different receptive fields are felt as separate
- Two points in one receptive field are felt as one
- This demonstrates the relationship between receptor density and discrimination



Lateral Inhibition

- **Lateral inhibition** (侧抑制) is a mechanism that sharpens sensory perception
- Central receptors in a stimulated area inhibit neighboring receptors
- This enhances contrast and improves stimulus localization

Lateral Inhibition

- When an object touches the skin (a), central receptors are most activated
- (b) Neighboring receptors are moderately activated
- (c) Lateral inhibition suppresses neighboring signals, sharpening perception
- This occurs in the central nervous system via interneurons



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Taste

- **Taste** (味觉) is mediated by **taste buds** (味蕾) on the tongue
- Each taste bud contains 50-100 **taste cells** (味觉细胞)
- Five primary tastes: salty (咸), sour (酸), sweet (甜), bitter (苦), umami (鲜)

Taste

- **Taste bud** structure:
- Chemicals dissolved at the pore bind to receptor proteins on microvilli
- Binding triggers neurotransmitter release
- This activates sensory neurons that transmit taste information



Taste

- Different taste mechanisms:
- **Salty**: Na^+ ions directly depolarize cells through ion channels
- **Sour**: H^+ ions depolarize cells
- **Sweet, umami, bitter**: G-protein-coupled receptors activate second messengers
- This diversity allows detection of various chemical properties



Smell

- **Olfaction** (嗅觉) detects airborne chemicals
- **Olfactory epithelium** (嗅上皮) contains millions of receptor neurons
- Humans have 400 different olfactory receptor types
- Each neuron expresses only one receptor type

Smell

- **Neural pathway for olfaction:**
- Olfactory receptor neurons → **olfactory bulb** (嗅球) → **glomeruli** (肾小球)
- **Tufted cells** (簇状细胞) and **mitral cells** (僧帽细胞) relay signals
- Signals travel to **olfactory cortex** (嗅皮层) in medial temporal lobes
- Each glomerulus receives input from only one receptor type



Smell

- Scanning electron micrograph of an **olfactory neuron** (嗅觉神经元)
- The tassel of **cilia** (纤毛) is clearly visible
- Cilia contain receptor proteins that bind odorant molecules
- This provides large surface area for detecting airborne chemicals



Smell

- Odorant signal transduction:
- Odorant binds receptor → G-protein (**transducin** 转导蛋白) activation
- **Adenylate cyclase** (腺苷酸环化酶) produces cAMP
- cAMP opens cation channels → $\text{Na}^+ / \text{Ca}^{2+}$ influx
- Depolarization generates action potentials



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Structure of Vestibular Apparatus

- **Vestibular apparatus** (前庭器官) detects head position and movement
- Components:
 - **Utricle** (椭圆囊) and **sacculle** (球囊): **Otolith organs** (耳石器官)
 - Three **semicircular canals** (半规管): Detect rotation
- Base of each canal has an **ampulla** (壶腹) with sensory hair cells

Structure of Vestibular Apparatus

- The **cochlea** (耳蜗) and **vestibular apparatus** form the inner ear
- Utricle and saccule detect linear acceleration and gravity
- Three semicircular canals detect rotational movements in three dimensions
- All structures contain sensory **hair cells** (毛细胞)



Structure of Vestibular Apparatus

- **Membranous labyrinth** (膜迷路) is filled with **endolymph** (内淋巴)
- **Bony labyrinth** (骨迷路) contains the membranous labyrinth
- Space between them is filled with **perilymph** (外淋巴)
- This dual-chamber system is essential for mechanotransduction



Sensory Hair Cells

- **Hair cells** are mechanoreceptors in the vestibular apparatus
- Each has **stereocilia** (静纤毛) and one **kinocilium** (动纤毛)
- Bending toward kinocilium → depolarization
- Bending away from kinocilium → hyperpolarization

Sensory Hair Cells

- Scanning electron micrograph of vestibular hair cells
- Multiple stereocilia of graduated heights
- Deflection of stereocilia opens mechanically-gated ion channels
- This converts mechanical stimulus into electrical signal



Otolith Organs

- **Utricle** and **sacculle** detect linear acceleration and head tilt
- Hair cells embedded in gelatinous membrane
- **Otoliths** (耳石): Calcium carbonate crystals that add mass
- Gravity or acceleration shifts otoliths, bending hair cells

Otolith Organs

- (a) Head upright: Otolith weight applies direct pressure
- (b) Head tilted: Otoliths shift, bending hair cells in specific direction
- Utricle: Sensitive to horizontal movements
- Saccule: Sensitive to vertical movements



Semicircular Canals

- Three canals oriented in different planes detect 3D rotation
- Each canal's ampulla contains a **cupula** (顶帽)
- Hair cells embedded in cupula
- Endolymph movement during rotation bends cupula

Semicircular Canals

- (a) At rest or constant velocity: Cupula and hair cells undisturbed
- (b) During rotation: Endolymph inertia causes relative movement
- Cupula bends, stimulating hair cells
- This signals angular acceleration



Semicircular Canals

- Vestibular information integrates in **vestibular nuclei** (前庭核)
- Also projects to **cerebellum** (小脑)
- These centers coordinate balance and posture
- Control eye movements to maintain gaze stability (**vestibulo-ocular reflex** 前庭眼反射)



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Ear Structure

- Three regions: **outer ear** (外耳), **middle ear** (中耳), **inner ear** (内耳)
- **Outer ear**: Pinna and auditory canal
- **Middle ear**: Ossicles (malleus, incus, stapes)
- **Inner ear**: Cochlea and vestibular apparatus

Ear Structure

- Sound pathway: Outer ear collects sound → tympanic membrane vibrates
- Middle ear ossicles amplify vibrations (20-fold)
- **Oval window** (卵圆窗) transmits to cochlea
- **Round window** (圆窗) allows pressure relief



Middle Ear

- Three **ossicles** (听小骨): **malleus** (锤骨), **incus** (砧骨), **stapes** (镫骨)
- **Tensor tympani** (鼓膜张肌) and **stapedius** (镫骨肌) muscles
- These muscles contract reflexively to loud sounds, protecting inner ear
- **Eustachian tube** (咽鼓管) equalizes air pressure

Middle Ear

- Medial view showing middle ear cavity
- Ossicles form mechanical lever system
- Muscles attach to ossicles for dampening
- Connection to pharynx maintains pressure balance



Cochlea Structure

- **Cochlea** (耳蜗) is spiral-shaped organ for hearing
- Three fluid-filled chambers:
 - **Scala vestibuli** (前庭阶): Perilymph
 - **Scala media** (中阶): Endolymph, contains organ of Corti
 - **Scala tympani** (鼓阶): Perilymph
- Chambers connect at apex via **helicotrema** (蜗孔)

Cochlea Structure

- Cross-section showing three turns of cochlea
- **Scala vestibuli** connects to oval window
- **Cochlear duct** (scala media) is between the other two
- **Scala tympani** connects to round window
- **Basilar membrane** (基底膜) separates scala media from scala tympani



Pitch Discrimination

- **Place theory** (位置理论): Different frequencies activate different cochlear regions
- High frequencies (20,000 Hz): Base of cochlea (stiff, narrow basilar membrane)
- Low frequencies (500 Hz): Apex of cochlea (flexible, wide basilar membrane)
- This creates a **tonotopic map** (音调定位图)

Pitch Discrimination

- Sound frequency determines location of maximum basilar membrane vibration
- Low pitch causes peak vibration near apex
- High pitch causes peak vibration near base
- This spatial coding allows frequency discrimination



Spiral Organ (Organ of Corti)

- **Spiral organ** (螺旋器) is the functional unit of hearing
- Contains **hair cells** (毛细胞): Inner (one row) and outer (three rows)
- **Tectorial membrane** (盖膜) overlies hair cells
- **Basilar membrane** (基底膜) supports the organ

Spiral Organ (Organ of Corti)

- (a) Organ of Corti within cochlear duct
- (b) Detailed view of hair cells and tectorial membrane
- Basilar membrane vibration → hair cell stereocilia bend → depolarization
- Inner hair cells: Primary sensory receptors
- Outer hair cells: Amplify vibrations



Spiral Organ (Organ of Corti)

- Tonotopic organization of the cochlea
- Different portions detect specific frequency ranges
- Base: High frequencies (narrower, stiffer region)
- Apex: Low frequencies (wider, more flexible region)
- This spatial frequency map is preserved in auditory pathways



Neural Pathways for Hearing

- Auditory pathway:
- Hair cells → auditory nerve (CN VIII) → **cochlear nuclei** (耳蜗核)
- → **superior olivary nucleus** (上橄榄核) → **inferior colliculus** (下丘)
- → **medial geniculate nucleus** (内侧膝状体) → **auditory cortex** (听觉皮层)

Neural Pathways for Hearing

- Neural pathway maintaining tonotopic organization
- Bilateral projections allow sound localization
- Multiple relay stations process different auditory features
- Primary auditory cortex in temporal lobe



Neural Pathways for Hearing

- Correlation between pitch and cortical location
- Auditory cortex maintains tonotopic map
- High frequencies represented in certain cortical areas
- Low frequencies in other areas
- Parallel to cochlear organization



Neural Pathways for Hearing

- The eye structure (overview for vision section)
- Complex optical system with multiple refractive surfaces
- **Cornea** (角膜) and **lens** (晶状体) focus light
- **Retina** (视网膜) contains photoreceptors



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Eye Structure

- Major eye structures:
 - **Cornea** (角膜): Primary refractive surface
 - **Iris** (虹膜): Controls pupil diameter
 - **Lens** (晶状体): Fine-tunes focus
 - **Retina** (视网膜): Light-sensitive layer
 - **Aqueous humor** (房水): Anterior chamber fluid
 - **Vitreous humor** (玻璃体): Posterior chamber gel

Eye Structure

- Internal anatomy of the eyeball
- Light enters from the right, passes through cornea and lens
- Image focused on retina at back of eye
- **Optic disc** (视盘): Blind spot where optic nerve exits
- **Fovea centralis** (中央凹): Region of highest visual acuity



Pupillary Control

- **Iris** contains two muscle types:
 - **Radial muscles** (放射状肌): Dilate pupil (sympathetic control)
 - **Circular muscles** (环形肌): Constrict pupil (parasympathetic control)
- **Pupillary reflex** (瞳孔反射) adjusts light entry

Pupillary Control

- Dim light: Radial muscles contract → pupil dilates
- Bright light: Circular muscles contract → pupil constricts
- This regulates retinal illumination
- Protects photoreceptors from damage and optimizes vision



Aqueous Humor

- **Aqueous humor** (房水) maintains **intraocular pressure** (眼内压)
- Produced by **ciliary body** (睫状体)
- Drains through **canal of Schlemm** (施莱姆管)
- Imbalance can cause **glaucoma** (青光眼)

Aqueous Humor

- Production and drainage pathway
- Aqueous humor circulates through anterior chamber
- Proper drainage maintains normal pressure
- Blocked drainage → elevated pressure → optic nerve damage



Aqueous Humor

- **Ophthalmoscope** (检眼镜) view of retina
- (a) Photograph and (b) illustration
- **Optic disc** (视盘): Blind spot, no photoreceptors
- Blood vessels radiate from optic disc
- **Macula lutea** (黄斑) contains fovea centralis



Refraction and Image Formation

- **Refraction** (折射): Bending of light at medium boundaries
- Cornea provides 70% of refractive power
- Lens provides 30% (adjustable)
- Light must converge to focus on retina

Refraction and Image Formation

- Image is inverted on retina
- Top of visual field projects to bottom of retina
- Bottom projects to top
- Brain interprets inverted image as upright



Refraction and Image Formation

- Image is also reversed left-to-right
- Right visual field projects to left side of retina
- Left field to right side
- Brain corrects for this reversal



Accommodation

- **Accommodation** (调节): Adjustment of lens shape for focusing
- **Ciliary muscle** (睫状肌) controls lens shape via **zonular fibers** (悬韧带)
- For near vision: Ciliary muscle contracts → lens more spherical
- For far vision: Ciliary muscle relaxes → lens flattens

Accommodation

- (a) Diagram of ciliary muscle and lens relationship
- (b) Scanning electron micrograph showing zonular fibers
- Fibers suspend lens and transmit ciliary muscle tension
- **Parasympathetic** control via **oculomotor nerve** (动眼神经)



Accommodation

- Changes in lens shape during accommodation
- Distant object: Lens flattened, less refractive power
- Near object: Lens rounded, greater refractive power
- **Presbyopia** (老花眼): Age-related loss of accommodation



Refractive Errors

- Common vision problems:
 - **Emmetropia** (正视): Normal vision
 - **Myopia** (近视): Nearsightedness, light focuses before retina
 - **Hyperopia** (远视): Farsightedness, light focuses behind retina
 - **Astigmatism** (散光): Irregular corneal curvature

Refractive Errors

- (a) Normal eye: Light focuses on retina
- (b) Myopia: Corrected with concave (diverging) lens
- (c) Hyperopia: Corrected with convex (converging) lens
- Lens power measured in **diopters** (屈光度)



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Retinal Structure

- **Retina** (视网膜) has three main cell layers:
 - **Ganglion cell layer** (神经节细胞层): Output neurons
 - **Bipolar cell layer** (双极细胞层): Intermediate neurons
 - **Photoreceptor layer** (感光细胞层): Rods and cones
- Light must pass through ganglion and bipolar cells to reach photoreceptors

Retinal Structure

- Layered organization of retina
- Light enters from left (through inner layers)
- Photoreceptors are in outermost layer
- This “inverted” arrangement is unique to vertebrate eyes
- **Horizontal** (水平细胞) and **amacrine cells** (无长突细胞) modulate signals



Photoreceptors

- Two types of photoreceptors:
 - **Rods** (视杆细胞): 120 million, dim light/night vision, no color
 - **Cones** (视锥细胞): 6 million, bright light/day vision, color vision
- Rods: Peripheral retina, high sensitivity
- Cones: Concentrated in fovea, high acuity

Photoreceptors

- Structural comparison of rods and cones
- Both have outer segment with photopigment
- Inner segment contains organelles
- Synaptic terminal releases neurotransmitter (glutamate)
- Different shapes optimize for different functions



Photopigments

- **Rhodopsin** (视紫红质) in rods consists of:
 - **Opsin** (视蛋白): Protein component
 - **Retinal** (视黄醛): Vitamin A derivative (chromophore)
- In darkness: 11-cis-retinal
- Light → isomerization to all-trans-retinal

Photopigments

- Rod discs contain rhodopsin molecules
- Thousands of discs in outer segment
- High concentration allows single photon detection
- Disc membranes constantly renewed



Photopigments

- **Photodissociation** (光解离) or bleaching of rhodopsin
- Light causes 11-cis-retinal → all-trans-retinal isomerization
- Conformational change in opsin
- Activates signal transduction cascade
- **Dark adaptation** (暗适应): Rhodopsin regeneration



Phototransduction

- In darkness:
 - **Dark current** (暗电流): Na^+ influx through cGMP-gated channels
 - Rod partially depolarized
 - Continuous glutamate release

Phototransduction

- (1) In dark: Na^+ enters, producing dark current
- (2) Rod partially depolarized
- (3) Continuous neurotransmitter release
- Light stops this process by closing Na^+ channels



Phototransduction

- In light:
- Activated rhodopsin → **transducin** (转导蛋白) activation
- → **Phosphodiesterase** (磷酸二酯酶) activation
- → cGMP hydrolysis → channels close
- → Hyperpolarization → reduced glutamate release



Color Vision

- Three cone types, each with different photopigment:
 - **S-cones** (短波锥体): Blue light (420 nm)
 - **M-cones** (中波锥体): Green light (530 nm)
 - **L-cones** (长波锥体): Red light (560 nm)
- **Trichromatic theory** (三色学说): Color from comparing cone activations

Color Vision

- Absorption spectra of three cone types
- S-cones peak at 420 nm (blue)
- M-cones peak at 530 nm (green)
- L-cones peak at 560 nm (red)
- Color perception from relative activation ratios



Visual Acuity

- **Fovea centralis** (中央凹) provides highest acuity
- Contains only cones (no rods)
- Minimal convergence (1:1 cone-to-ganglion ratio)
- Lateral cell displacement allows direct light access

Visual Acuity

- When eyes “track” an object, image falls on fovea
- Central vision uses foveal cones
- Peripheral vision uses rods
- Eye movements constantly reposition fovea on objects of interest



Visual Acuity

- **Convergence** (会聚) in the retina
- Fovea: Low convergence → high acuity, low sensitivity
- Periphery: High convergence (many rods → one ganglion) → low acuity, high sensitivity
- Trade-off between resolution and light sensitivity



Visual Pathways

- Visual pathway:
- Photoreceptors → bipolar cells → ganglion cells → optic nerve
- → **Optic chiasm** (视交叉) → **optic tract** (视束)
- → **Lateral geniculate nucleus** (外侧膝状体)
- → **Optic radiations** (视辐射) → **Primary visual cortex** (初级视觉皮层)

Visual Pathways

- At optic chiasm: Nasal retinal fibers cross
- Temporal retinal fibers remain ipsilateral
- Result: Each hemisphere processes contralateral visual field
- Left hemisphere: Right visual field
- Right hemisphere: Left visual field



Visual Pathways

- **Striate cortex** (纹状皮层) or **area 17**:
Primary visual cortex
- Located in occipital lobe
- Processes basic visual features
- Projects to **extrastriate areas** (纹外区域) for higher processing



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Receptive Fields

- **Ganglion cell receptive fields** (感受野):
 - **On-center cells** (开中心细胞): Excited by light in center
 - **Off-center cells** (关中心细胞): Inhibited by light in center
- Surround has opposite response
- This enhances contrast and edge detection

Receptive Fields

- Center-surround organization
- On-center: Light in center → excitation; light in surround → inhibition
- Off-center: Opposite pattern
- Enhances detection of borders and contrast
- Created by lateral inhibition via horizontal and amacrine cells



Cortical Processing

- Visual cortex contains specialized cell types:
 - **Simple cells** (简单细胞): Respond to oriented edges
 - **Complex cells** (复杂细胞): Respond to moving edges
 - **Hypercomplex cells** (超复杂细胞): Respond to corners and angles

Cortical Processing

- **Simple cells** in primary visual cortex
- Respond to bars or edges at specific orientations
- Different cells tuned to different angles
- Hierarchical processing builds complex feature detection
- Foundation of object recognition



Visual Processing Streams

- Two major processing pathways:
 - **Dorsal stream** (背侧流): “Where pathway” - spatial location and motion
 - **Ventral stream** (腹侧流): “What pathway” - object identification and color
- Dorsal: V1 → parietal cortex
- Ventral: V1 → temporal cortex
- Parallel, hierarchical processing